

ABE 721

Agricultural Systems Modelling and Simulation

Module 5: Simulating Materials Flow

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Chapter 5

Simulating Materials Flow

Agricultural systems are fundamentally thermodynamic and biogeochemical engines. They do not generate mass from nothing; they receive exogenous inputs of energy (solar radiation) and utilize that energy to physically reorganize and transport chemical materials (carbon, nitrogen, water, phosphorus) from the environment into harvestable biological structures.

As extensively detailed in the texts of Haefner (1996) and Peart & Curry (1998) within your syllabus, understanding the physics and biology of an agricultural system explicitly requires simulating the **Flow of Materials**. This module explores the mathematical architectures used to track the continuous, cyclical transfer of elements between interconnected ecosystem compartments. It further examines how to integrate these flows across strict spatial boundaries and temporal scales, mathematically models the evaporative phase change of water flows, and finally simulates the active management interventions (like fertilizer application and harvest) that disrupt and redirect these natural flows.

5.1 5.1 Simulating Cyclical Systems

In natural and agricultural ecology, elements such as carbon (C), nitrogen (N), and phosphorus (P) are never destroyed; they are continually recycled between the atmosphere, the soil matrix, microbial biomass, and plant tissue. Simulating these cyclical systems requires the application of **Compartment Modelling**.

5.1.1 Fundamentals of Compartment Modelling

A compartment model strictly divides a continuous physical ecosystem into a finite number of discrete, interconnected "pools" or "stocks" of material. The mathematical simulation relies absolutely on the fundamental principle of the **Conservation of Mass**.

Let $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$ be the state vector representing the mass of a specific material (e.g., Nitrogen in kg/ha) existing simultaneously in n distinct compartments. The transient rate of change of mass in any specific compartment i is calculated by summing all mass flows entering from other compartments j, subtracting all mass flows exiting to other compartments, adding external environmental inputs (z_i), and subtracting external exports to the environment (y_i):

$$\frac{dx_i}{dt} = \sum_{j \neq i} F_{ji} - \sum_{j \neq i} F_{ij} + z_i - y_i \tag{5.1}$$

Where F_{ij} represents the physical mass flow rate from compartment i to compartment j.

Linear Donor-Controlled Systems

In the majority of biogeochemical models, the flow rate F_{ij} is assumed to be biologically or physically proportional to the amount of material currently existing in the source (donor) compartment i. This is known as a linear donor-controlled flow:

$$F_{ij} = a_{ij} x_i \tag{5.2}$$

Where a_{ij} is the fractional transfer coefficient (measured in $time^{-1}$). This coefficient mathematically defines the fraction of compartment i that is physically transferred to compartment j per unit of time.

If an agricultural ecosystem is entirely donor-controlled, the entire complex cycle of nutrient transfer can be elegantly formulated as a massive system of linear

differential equations using matrix algebra:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{z}(t) \tag{5.3}$$

Where **A** is the extremely dense compartmental transfer matrix. The diagonal elements a_{ii} represent the total loss rate from compartment i (i.e., $a_{ii} = -\sum_{j \neq i} a_{ij} - a_{i0}$), and the off-diagonal elements a_{ij} represent the specific gain to i strictly from j.

5.1.2 Residence Time and Turnover Rates

A critical analytical metric in cyclical material simulation is the **Mean Residence Time** (τ_i), which calculates the statistically average duration a single atom of material (e.g., a Carbon atom) spends inside compartment i before being biologically metabolized or leached out.

In a steady-state equilibrium ecosystem (where $\frac{dx_i}{dt} = 0$), the absolute Total Inflow equals the absolute Total Outflow ($F_{out,i}$). The mean residence time is simply calculated as:

$$\tau_i = \frac{x_i^*}{F_{out,i}} = \frac{1}{|a_{ii}|} \tag{5.4}$$

The reciprocal of the residence time ($1/\tau_i$) is the **Turnover Rate**, which dictates how rapidly the compartment flushes and replaces its internal mass. For instance, the turnover rate of microbial biomass nitrogen in soil is measured in days, whereas the turnover rate of stable humic soil organic matter is measured mathematically in centuries.

5.1.3 Case Study: Simulating the Carbon Cycle in Soil

To physically demonstrate cyclical simulation, consider the Rothamsted Carbon Model (RothC), a globally utilized framework for simulating soil organic carbon (SOC) turnover.

The model divides total soil carbon into multiple compartments based strictly on their biological decomposition resistance: 1. DPM: Decomposable Plant Material

(Fast turnover) 2. RPM: Resistant Plant Material (Medium turnover) 3. BIO: Microbial Biomass (Fast turnover recycler) 4. HUM: Humified Organic Matter (Slow turnover)

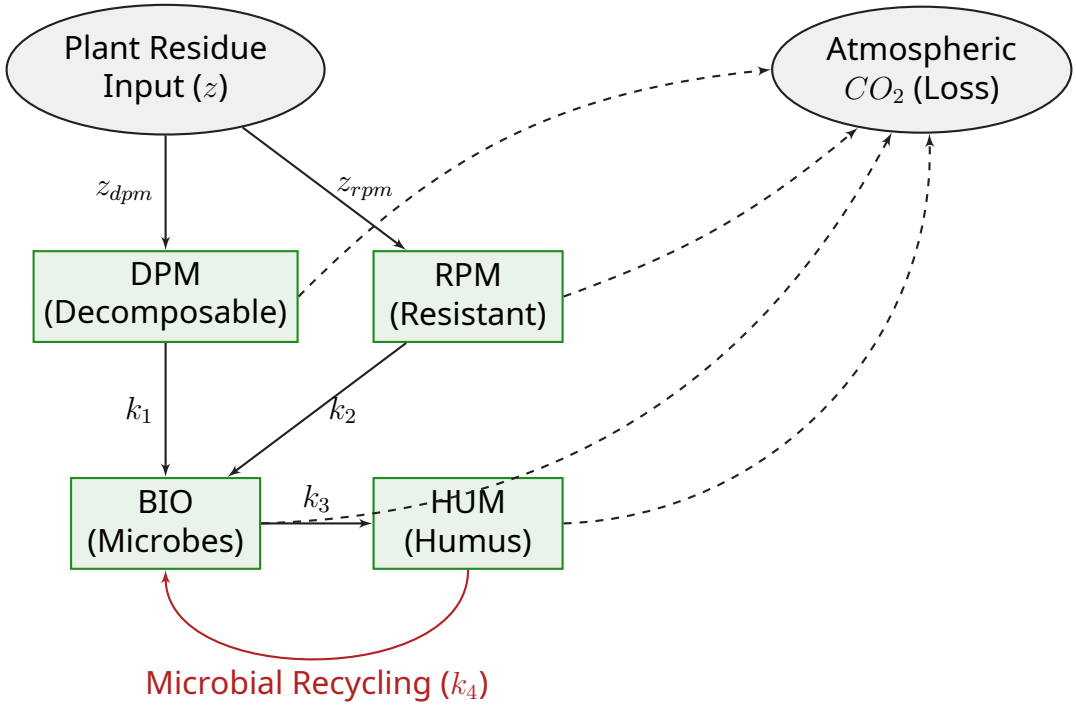


Figure 5.1: Compartment flow simulation of the Soil Carbon Cycle. Solid arrows represent mass transfers between internal matrix states, while dashed arrows represent first-order respiratory losses from the system boundaries.

The cyclical nature arises because as microbes (BIO) decompose humus (HUM), they actually die and their physical mass cycles backward to become new organic input, creating an infinite, decaying feedback loop of material flow. This is mathematically defined by a system of ODEs:

$$\frac{d(BIO)}{dt} = a \cdot k_1(DPM) + b \cdot k_2(RPM) + c \cdot k_4(HUM) - k_3(BIO) \tag{5.5}$$

Where k_i are the specific, temperature-dependent decomposition rate constants, and a, b, c represent the precise fraction of decomposed carbon that is physically assimilated by microbes rather than respired instantly as CO_2 gas.

5.2 5.2 Spatial and Temporal Models Integration

As explicitly discussed in Module 2, Lumped (0D) models treat an entire farm field as a single, perfectly mixed homogenous bucket. However, the physical flow of agricultural materials—such as the massive leaching of highly soluble nitrate fertilizers, or the spread of an aggressive nematode infestation—occurs simultaneously across both geographic space and continuous time. Simulating these materials explicitly requires the integration of spatial algorithms and temporal calculus.

5.2.1 The Mathematics of Spatio-Temporal Transport

The movement of a dissolved material (a solute, like fertilizer or pesticide) flowing through a porous agricultural soil matrix is universally governed by the **Advection-Dispersion-Reaction Equation (ADRE)**. This is one of the most rigorously complex Partial Differential Equations (PDEs) in all of agricultural systems engineering.

Derivation of the ADRE

Consider a 1-dimensional control volume of soil. The transient change in the concentration of the solute $C(x,t)$ within that volume is driven by three distinct, simultaneous physical processes: 1. **Advection:** The physical transport of the solute mass strictly due to the bulk flow of the carrier fluid (water). The advective flux is $J_a = v \cdot C$, where v is the average pore water velocity. 2. **Dispersion/Diffusion:** The transport of solute mass due to microscopic concentration gradients and tortuous mechanical mixing in the soil pores. Governed by Fick's Law, the dispersive flux is $J_d = -D\frac{\partial C}{\partial x}$, where D is the hydrodynamic dispersion coefficient. 3. **Reaction/Decay:** The biological or chemical transformation of the solute mass within the matrix (e.g., microbial denitrification or chemical half-life degradation). This is generally modelled as a first-order decay sink: $R = -\mu C$.

By rigorously applying the principle of the Conservation of Mass across the geometric boundaries of the control volume, we sum the spatial derivatives of the

fluxes to derive the complete, transient 1D ADRE:

$$\frac{\partial C}{\partial t} = \underbrace{D\frac{\partial^2 C}{\partial x^2}}_{\text{Dispersion}} - \underbrace{v\frac{\partial C}{\partial x}}_{\text{Advection}} - \underbrace{\mu C}_{\text{Reaction}} \tag{5.6}$$

For 2D spatial simulations (e.g., fertilizer leaching both downwards into aquifers and laterally into adjacent streams), the equation expands massively into multivariate calculus:

$$\frac{\partial C}{\partial t} = \left[D_x\frac{\partial^2 C}{\partial x^2} + D_z\frac{\partial^2 C}{\partial z^2}\right] - \left[v_x\frac{\partial C}{\partial x} + v_z\frac{\partial C}{\partial z}\right] - \mu C \tag{5.7}$$

5.2.2 Numerical Integration: The ADI Method

Solving the 2D ADRE analytically is impossible for real-world heterogeneous farms. It must be solved computationally using a discrete spatio-temporal grid. However, standard Explicit Finite Difference Methods (FDM) suffer from severe, catastrophic Courant-Friedrichs-Lewy (CFL) stability constraints when handling advection-dominated flows. If the fluid velocity v is high, the simulation mathematically explodes.

To solve this, engineers integrate temporal and spatial dimensions using the **Alternating Direction Implicit (ADI) Method**. ADI is a highly advanced, unconditionally stable computational algorithm that effectively splits a massive 2D PDE grid problem into two much simpler 1D problems, alternating the direction of implicit calculation at every half time-step.

Step 1 (X-Sweep): Advance the simulation exactly half a time-step ($t + \Delta t/2$). Calculate the X-derivatives *implicitly* (meaning future states depend on other unknown future states), while calculating the Z-derivatives *explicitly* (using known past states).

$$\frac{C_{i,k}^{n+1/2} - C_{i,k}^{n}}{\Delta t/2} = D_x\delta_x^2(C^{n+1/2}) + D_z\delta_z^2(C^n) - \text{Advection terms}\ldots \tag{5.8}$$

This forms a massive tridiagonal matrix system exclusively along the horizontal X-rows, which can be solved instantly using the Thomas Algorithm.

Step 2 (Z-Sweep): Advance the simulation the final half time-step ($t + \Delta t$). Now, calculate the Z-derivatives *implicitly* and use the results from Step 1 explicitly for the X-derivatives.

$$\frac{C_{i,k}^{n+1} - C_{i,k}^{n+1/2}}{\Delta t/2} = D_x \delta_x^2(C^{n+1/2}) + D_z \delta_z^2(C^{n+1}) - \text{Advection terms} \ldots \quad (5.9)$$

This forms independent tridiagonal systems exclusively along the vertical Z-columns.

By rapidly alternating between horizontal and vertical sweeps, the ADI computer algorithm can simulate the complex 2D flow of toxic pesticides through thousands of meters of soil over several years in a matter of seconds, without ever violating mathematical stability constraints.

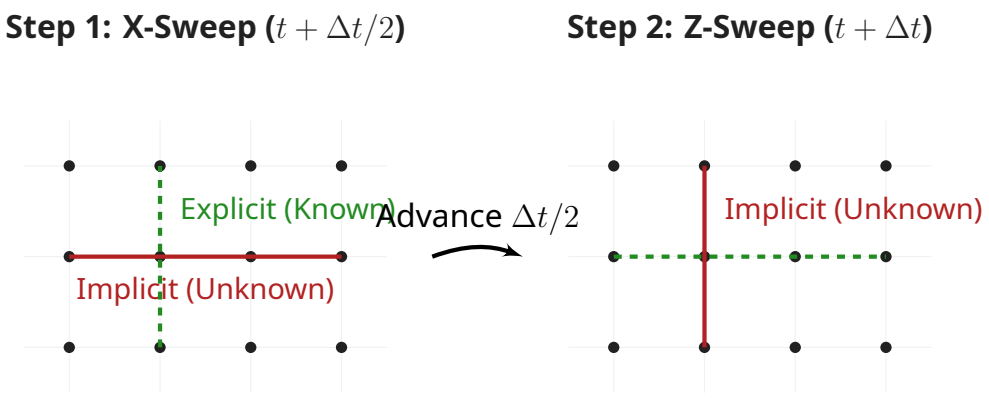


Figure 5.2: The Alternating Direction Implicit (ADI) method for integrating spatial PDEs over time. By mathematically splitting the 2D spatial grid into alternating 1D sweeps, engineers can simulate complex material flows (like fertilizer plumes) with absolute numerical stability.

5.3 5.3 Evaporation Models and Atmospheric Flows

Simulating the flow of materials in an agricultural system is not strictly limited to solids and liquids. The most massive and energetic material transfer occurring on any farm is the continuous phase-change flow of liquid water from the soil and crop canopy into atmospheric vapor. This is evaporation.

As heavily referenced by Kramer (2012) and Michael (2008), accurately simulating the evaporation rate (E) is critical for modeling irrigation demands, reservoir depletion, and global watershed hydrology. Mathematical models of evaporation fundamentally divide into three historical paradigms.

5.3.1 1. Mass Transfer (Aerodynamic) Models

Originating from Dalton's law of partial pressures (1802), mass transfer models treat evaporation strictly as an aerodynamic diffusion process. The upward flow of water vapor is driven exclusively by the difference in vapor pressure between the saturated evaporating surface (e_s) and the ambient air above it (e_a).

The general mathematical formulation is:

$$E = f(u) \cdot (e_s - e_a) \tag{5.10}$$

Where $f(u)$ is a complex wind function representing aerodynamic turbulence, and $(e_s - e_a)$ is the Vapor Pressure Deficit (VPD).

In modern systems engineering, this is formalized using an **Aerodynamic Resistance** (r_a) term. For a neutral atmospheric boundary layer, the resistance to vapor flow is integrated logarithmically based on wind speed (u_z) measured at height z:

$$r_a = \frac{\ln\left(\frac{z-d}{z_{om}}\right) \ln\left(\frac{z-d}{z_{oh}}\right)}{k^2 u_z} \tag{5.11}$$

Where d is zero-plane displacement height, z_{om} is the roughness length for momentum, and k is the von Karman constant (0.41). A rough, tall crop canopy like sugarcane creates intense mechanical turbulence, massively lowering r_a and ac-

celerating the upward flow of water material.

5.3.2 2. Energy Balance Models

Water cannot simply evaporate because the air is dry; it absolutely requires massive inputs of thermal energy to break hydrogen bonds. The latent heat of vaporization (λ) is approximately 2.45 MJ per kg of water.

Energy balance models treat the field strictly as a thermodynamic control volume. The First Law of Thermodynamics mandates that net incoming solar radiation (R_n) must equal the sum of the outgoing Sensible Heat Flux (H) warming the air, the Latent Heat Flux (λE) evaporating the water, and the Soil Heat Flux (G).

$$R_n = H + \lambda E + G \tag{5.12}$$

By utilizing the **Bowen Ratio** ($\beta = H/\lambda E$), which relates sensible to latent heat, engineers can mathematically eliminate the difficult-to-measure sensible heat flux (H) entirely:

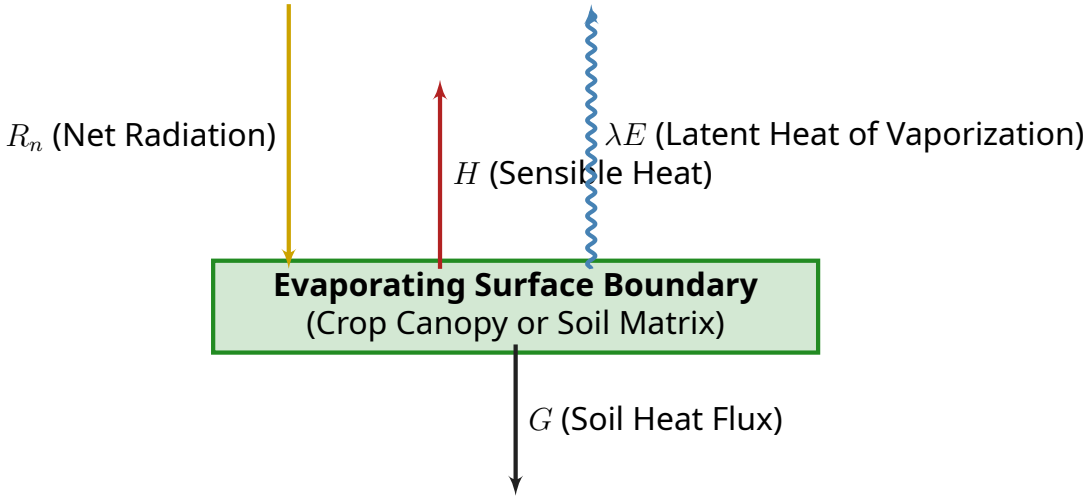
$$R_n = \beta \lambda E + \lambda E + G \implies \lambda E = \frac{R_n - G}{1 + \beta} \tag{5.13}$$

This formulation proves that the absolute theoretical maximum rate of material evaporation is strictly dictated by the available radiant solar energy.

5.3.3 3. The Combination Method: Penman-Monteith

The aerodynamic models fail when radiation is low, and the energy balance models fail when strong dry winds advect heat from adjacent deserts (the oasis effect). The legendary physicist Howard Penman brilliantly combined both physics into a single, unified differential equation. John Monteith later added the biological **Canopy Surface Resistance** (r_s) to account for the physical stomata of plants restricting flow.

The resulting **FAO-56 Penman-Monteith Equation** is the absolute global standard



Conservation: $R_n - G - H - \lambda E = \frac{dU}{dt} \approx 0$

Figure 5.3: The thermodynamic Energy Balance driving material evaporation. The incoming radiation (R_n) must provide the massive 2.45 MJ/kg of energy required to force the phase change of liquid water into atmospheric vapor (λE).

for simulating the flow of evaporated materials in agricultural systems:

$$\lambda E T_0 = \frac{\Delta(R_n - G) + \rho_a c_p \frac{(e_s - e_a)}{r_a}}{\Delta + \gamma \left(1 + \frac{r_s}{r_a}\right)} \tag{5.14}$$

Where Δ is the strict mathematical slope of the saturation vapor pressure curve, acting as a weighting factor between the aerodynamic term (right) and the radiation term (left), heavily favoring radiation at high temperatures.

When simulating *bare soil evaporation*, r_s mathematically approaches infinity as the top layer of soil dries, instantly severing the hydraulic continuity and violently halting the upward flow of water material, protecting deep subsoil moisture.

5.4 5.4 Crop Growth and Active Management Simulation

Natural ecosystems simulate material flows reaching climax equilibrium over centuries. Agricultural systems, however, are violently and actively managed. Humans continuously inject massive exogenous impulses of materials (fertilizers, irrigation) and forcibly sever biological flows (harvesting). Simulating crop growth requires integrating these aggressive, discrete management interventions into the continuous biological differential equations.

5.4.1 Resource-Limited Biomass Accumulation

As detailed in Peart & Curry (1998), potential crop growth assumes perfect conditions. Real-world simulated growth is strictly limited by the most deficient material flow, adhering mathematically to **Liebig's Law of the Minimum**.

The actual daily biomass growth rate ($\frac{dW}{dt}_{act}$) is the potential maximum rate multiplied by a dimensionless scalar stress factor ($0 \leq Stress \leq 1$):

$$\frac{dW}{dt}_{act} = \frac{dW}{dt}_{pot} \times \min(F_{water}, F_{nitrogen}, F_{temp}) \tag{5.15}$$

Water Stress Factor (F_{water})

Water stress restricts the flow of carbon. It is calculated dynamically based on the ratio of actual transpiration (T_{act}, limited by soil moisture conductivity) to potential transpiration (T_{pot}, dictated by Penman-Monteith atmospheric demand):

$$F_{water} = \frac{T_{act}}{T_{pot}} \tag{5.16}$$

When the soil dries and hydraulic resistance skyrockets, T_{act} drops below T_{pot}, driving F_{water} below 1.0, which instantly cripples the $\frac{dW}{dt}$ calculus, simulating drought stunting.

5.4.2 Simulating Management: Discrete Impulses in Continuous Time

How does a computer algorithm simulate a tractor driving through a field applying 100 kg/ha of Urea fertilizer on precisely Day 45?

Because the crop growth PDEs run continuously, management events are modeled as **Dirac Delta Functions** ($\delta(t)$) or instantaneous state-resets. They are *discrete events* interrupting a *continuous simulation* (referencing the paradigms of Module 3.1).

1. Simulating Irrigation Events

Let $\theta(t)$ be the continuous volumetric soil moisture state variable. It decays continuously due to the ET ODEs. An irrigation event scheduled at time t_{irrig} with an application depth of I_{depth} is simulated as an instantaneous mathematical jump condition:

$$\theta(t_{irrig}^{+}) = \theta(t_{irrig}^{-}) + \frac{I_{depth}}{Z_{root}} \tag{5.17}$$

The algorithm then strictly checks if the new θ exceeds the soil's Field Capacity (FC). If it does, the excess material is instantly shunted into the deep drainage/leaching flow algorithms, simulating over-irrigation runoff.

2. Simulating Harvest Events

A harvesting event is the ultimate disruption of material flow. The accumulated grain biomass stock is instantly removed from the system boundary (yielding output Y), and the remaining vegetative biomass (leaves, stems) is instantly transferred into the DPM (Decomposable Plant Material) soil carbon compartment discussed in Section 5.1, explicitly restarting the biogeochemical cycle.

$$Y_{harvest} = W_{grain}(t_{harvest}^{-}) \tag{5.18}$$

$$DPM(t_{harvest}^{+}) = DPM(t_{harvest}^{-}) + W_{leaf}(t_{harvest}^{-}) + W_{stem}(t_{harvest}^{-}) \tag{5.19}$$

$$W_{total}(t_{harvest}^{+}) = 0 \quad \text{(Biological state reset to zero)} \tag{5.20}$$

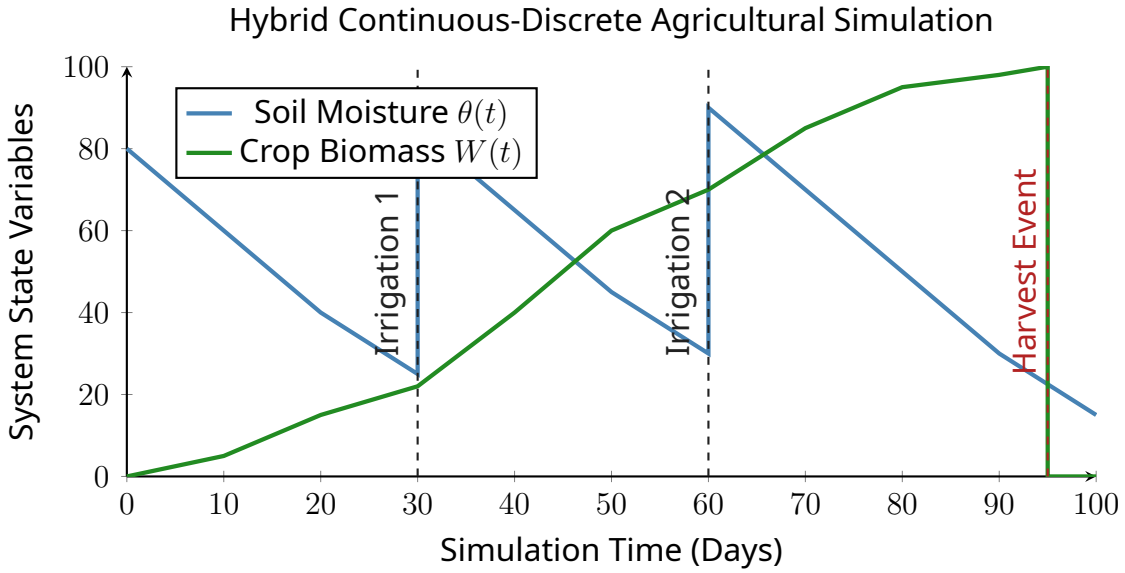


Figure 5.4: The execution of a hybrid systems simulation. The smooth curves represent the continuous numerical integration of ordinary differential equations (ODEs). The vertical discontinuities represent mathematically instantaneous discrete management events executed by the farmer, abruptly altering the flow of materials.

Through the rigorous mathematical integration of biogeochemical cyclical compartments (5.1), spatio-temporal PDE grids (5.2), phase-change thermodynamic evaporation constraints (5.3), and the impulse logic of human management (5.4), the modern agricultural engineer possesses the computational power to simulate, predict, and ultimately optimize the complete existence of any farming system on Earth.

End of Module 5.